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Drying machine

Abstract

A drying machine, comprising a box body. A first partition plate is provided in the box body, and the first partition plate is located in front of a rear wall of the box body; a drying cavity and a first accommodation cavity are respectively located on two sides of the first partition plate; the first accommodation cavity is defined between the first partition plate and the rear wall; a fan wheel is provided in the first accommodation cavity, and a rotation axis of the fan wheel is perpendicular to the first partition plate and the rear wall; a first air inlet is formed in the first partition plate, a second air inlet is formed in the rear wall; the first air inlet and the second air inlet are respectively opposite to the axial direction of the fan wheel; the fan wheel realize the inner circulation and outer circulation of air.

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Background/Summary

CROSS REFERENCES TO THE RELATED APPLICATIONS

(1) The application is a national stage entry of PCT/CN2021/119365 filed on Sep. 18, 2021, which claims priority to Chinese patent applications No. 202010992640.0 filed on Sep. 21, 2020, 202010992806.9 filed on Sep. 21, 2020, 202022070191.2 filed on Sep. 21, 2020, 202022086090.4 filed on Sep. 21, 2020, and 202022087091.0 filed on Sep. 21, 2020, the entire contents of which are incorporated herein by reference.

TECHNICAL FIELD

(2) The present disclosure relates to the technical field of drying apparatuses, and in particular to a drying machine.

BACKGROUND

(3) A pet drying box is an apparatus for drying the fur of a pet after the pet is bathed and groomed. In the related art, most of pet drying boxes are made by assembling simple fan wheels and heating assemblies, so that the pet drying boxes have no reasonable air channel design, resulting in poor drying effects and large noise and affecting the comfort of pets in the pet drying boxes.

Alternatively, the fan wheel of the pet drying box is arranged at the top or bottom of the pet drying box, wherein the arrangement of the fan wheel at the top of the pet drying box provides a longer air channel and causes a great loss of wind energy and low drying efficiency, and the arrangement of

the fan wheel at the bottom of the pet drying box may result in liquid leakage which causes a safety risk of the pet drying box.

SUMMARY

(4) The present disclosure aims to solve at least one of the technical problems existing in the prior art. For this purpose, an object of the present disclosure is to provide a drying machine which increases its air supply rate, reduces the loss of wind energy, improves its drying efficiency, reduces noise during its operation, and has a reduced production cost.

(5) The drying machine according to an embodiment of the present disclosure comprises: a box body, wherein a first partition plate is provided in the box body, and the first partition plate is located in front of a rear wall of the box body; a drying cavity and a first accommodation cavity are respectively located on two sides of the first partition plate; the first accommodation cavity is defined between the first partition plate and the rear wall; a fan wheel is provided in the first accommodation cavity, and a rotation axis of the fan wheel is perpendicular to the first partition plate and the rear wall; a first air inlet is formed in the first partition plate, a second air inlet is formed in the rear wall, and the first air inlet and the second air inlet each are axially opposite to the fan wheel; when the fan wheel rotates, an airflow inside the drying cavity is suitable for entering the first accommodation cavity through the first air inlet under the action of the fan wheel and then flowing into the drying cavity, and the airflow outside the drying machine is suitable for entering the first accommodation cavity through the second air inlet under the action of the fan wheel and then flowing into the drying cavity, so as to realize the inner circulation and the outer circulation of air by means of one fan wheel.

(6) According to the drying machine, such as the pet drying machine of the embodiment of the present disclosure, the interior of the box body is divided into the drying cavity and the first accommodation cavity by locating the first partition plate in front of the rear wall of the box body, the first accommodation cavity is defined between the first partition plate and the rear wall of the box body, the fan wheel is provided in the first accommodation cavity, the first air inlet is formed in the first partition plate, the second air inlet is formed in the rear wall, and the first air inlet and the second air inlet each are axially opposite to the fan wheel. Thus, the air supply rate of the drying machine is increased, the loss of wind energy is reduced, the drying efficiency of the drying machine is improved, the nonuniformity of the hot and cold air in the box body caused by an additional fan wheel can also be avoided, and the noise during the operation of the drying machine and the production cost of the drying machine can be reduced.

(7) According to some embodiments of the present disclosure, a fan wheel housing is provided between the first partition plate and the rear wall, the fan wheel is arranged in the fan wheel housing, and an air channel is defined between the fan wheel and the fan wheel housing, and when the fan wheel rotates, the airflow in the drying cavity sequentially flows through the first air inlet and the air channel and then flows into the drying cavity, and the airflow outside the drying machine sequentially flows through the second air inlet and the air channel and then flows into the drying cavity.

(8) According to some embodiments of the present disclosure, a second partition plate is provided in the box body, a second accommodation cavity in direct communication with an air channel outlet of the air channel is defined between the second partition plate and an inner wall of the box body at a position in the fan wheel housing corresponding to the air channel outlet, an air outlet is formed in the second partition plate, and the drying cavity is located above the second partition plate; and when the fan wheel operates to rotate, the airflow inside the drying cavity is suitable for sequentially flowing through the first air inlet, the air channel, the second accommodation cavity and flowing into the drying cavity through the air outlet, and the airflow outside the drying machine is suitable for sequentially flowing through the second air inlet, the air channel, the second accommodation cavity and flowing into the drying cavity through the air outlet.

(9) According to some embodiments of the present disclosure, the second partition plate is

removably arranged in a lower portion of the box body.

(10) According to some embodiments of the present disclosure, the air outlet comprises a plurality of air outlet orifices; and the second partition plate comprises: a partition plate body; and two partition plate sections, wherein the two partition plate sections are respectively arranged on the left side and the right side of the partition plate body, the two partition plate sections extend obliquely upwards from bottom to top in a direction away from the center of the partition plate body, and a plurality of air outlet orifices are formed in both the two partition plate sections and the partition plate body.

(11) According to some embodiments of the present disclosure, each partition plate section is an arc-shaped partition plate projecting away from the partition plate body. According to some embodiments of the present disclosure, a heating assembly is provided at an air outlet of the spiral air channel.

(12) According to some embodiments of the present disclosure, a negative ion generator is provided on the first partition plate, and an output end of the negative ion generator extends into the first accommodation cavity; and an ozone disinfection apparatus is provided on the heating assembly.

(13) According to some embodiments of the present disclosure, the air channel is a spiral air channel.

(14) According to some embodiments of the present disclosure, a projection of at least a portion of the fan wheel housing on the first partition plate is in the form of a spiral.

(15) According to some embodiments of the present disclosure, a projection of at least the portion of the fan wheel housing on the first partition plate is in the form of an Archimedean spiral.

(16) According to some embodiments of the present disclosure, the air channel has an air channel outlet, and the cross-sectional area of at least a portion of the air channel gradually increases toward the air channel outlet in a circumferential direction of the fan wheel.

(17) According to some embodiments of the present disclosure, the fan wheel housing is arranged on at least one of the first partition plate and the rear wall.

(18) According to some embodiments of the present disclosure, the drying box further comprises an electric motor support; and the fan wheel housing comprises: a first fan wheel housing arranged on the first partition plate; and a second fan wheel housing arranged on the electric motor support, the second fan wheel housing and the first fan wheel housing jointly defining a fan wheel accommodation cavity for accommodating the fan wheel.

(19) According to some embodiments of the present disclosure, the electric motor support is formed with an air inlet hole that runs therethrough in an axial direction of the fan wheel, the electric motor support is in sealed connection with the rear wall on a peripheral side of the air inlet hole, and outside air sucked by the fan wheel sequentially passes through the second air inlet and the air inlet hole and enters the fan wheel accommodation cavity.

(20) According to some embodiments of the present disclosure, the electric motor support is detachably connected to the first partition plate.

(21) According to some embodiments of the present disclosure, an electric motor is provided in the fan wheel accommodation cavity, an output shaft of the electric motor is fixed to the fan wheel, and the electric motor is mounted on the second fan wheel housing.

(22) According to some embodiments of the present disclosure, a rear surface of the first partition plate extends backwards from the first fan wheel housing.

(23) According to some embodiments of the present disclosure, the fan wheel is a forward-inclined fan wheel.

(24) According to some embodiments of the present disclosure, the fan wheel is provided as an axially through fan wheel.

(25) According to some embodiments of the present disclosure, a wall forming the drying cavity is formed with at least one through ventilation hole.

(26) According to some embodiments of the present disclosure, at least one cat stroking opening is formed in the box body, and a cat stroking door is provided at the cat stroking opening for opening or closing the cat stroking opening.

(27) According to some embodiments of the present disclosure, an opening is formed in the box body, a removable viewing window is provided at the opening, and the cat stroking opening is formed in the viewing window.

(28) According to some embodiments of the present disclosure, one end of the cat stroking door is pivotably connected to the viewing window by means of a pivoting assembly; and a locking assembly is provided between the other end of the cat stroking door and the viewing window, the cat stroking door is adapted for opening the cat stroking opening when the locking assembly is unlocked, and the cat stroking door is fixed to the viewing window when the other end of the cat stroking door is locked by the locking assembly.

(29) According to some embodiments of the present disclosure, a placement slot is formed in the box body, a fragrance box is removably arranged in the placement slot by means of a magnetic attraction assembly, the fragrance box has a first configuration and a second configuration in the placement slot, at least one through hole is formed in the fragrance box, the interior of the fragrance box is not in communication with the interior of the box body when the fragrance box is in the first configuration, and the interior of the fragrance box is in communication with the interior of the box body through the through hole when the fragrance box is in the second configuration.

(30) Some of additional aspects and advantages of the present disclosure will be given in the following description, and will be apparent from the following description, or may be learned by practice of the present disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The foregoing and/or additional aspects and advantages of the present disclosure will become apparent and easily understood from the following description of the embodiments taken in conjunction with the accompanying drawings, in which:
- (2) FIG. 1 is a schematic diagram of a drying machine according to an embodiment of the present disclosure;
- (3) FIG. 2 is an enlarged view of part A circled in FIG. 1;
- (4) FIG. 3 is a schematic diagram of the drying machine according to an embodiment of the present disclosure, in which a fan wheel housing and a fan are not shown;
- (5) FIG. 4 is an exploded view of the drying machine according to an embodiment of the present disclosure;
- (6) FIG. 5 is a schematic diagram of a fragrance box of the drying machine in a first configuration according to an embodiment of the present disclosure;
- (7) FIG. 6 is an enlarged view of part B circled in FIG. 5;
- (8) FIG. 7 is a schematic diagram of the fragrance box of the drying machine in a second configuration according to an embodiment of the present disclosure;
- (9) FIG. 8 is an enlarged view of part C circled in FIG. 7;
- (10) FIG. 9 is an enlarged view of part D circled in FIG. 7;
- (11) FIG. 10 is an enlarged view of part E circled in FIG. 7;
- (12) FIG. 11 is a cross-sectional view of the drying machine according to an embodiment of the present disclosure;
- (13) FIG. 12 is an enlarged view of part F circled in FIG. 11;
- (14) FIG. 13 is a cross-sectional view of the drying machine at another view angle according to an embodiment of the present disclosure;

- (15) FIG. 14 is a top view of a second partition plate of the drying machine according to an embodiment of the present disclosure;
- (16) FIG. 15 is a schematic diagram of the second partition plate of the drying machine according to an embodiment of the present disclosure;
- (17) FIG. 16 is an enlarged view of part G circled in FIG. 15;
- (18) FIG. 17 is a partial cross-sectional view of the drying machine according to an embodiment of the present disclosure;
- (19) FIG. 18 is an enlarged view of part H circled in FIG. 17;
- (20) FIG. 19 is an enlarged view of part I circled in FIG. 17;
- (21) FIG. 20 is a schematic diagram of a cat stroking door and a viewing window of the drying machine according to an embodiment of the present disclosure;
- (22) FIG. 21 is a side view of a partial structure of the viewing window according to an embodiment of the present disclosure;
- (23) FIG. 22 is a schematic diagram showing the cat stroking door and the viewing window of the drying machine at another view angle according to an embodiment of the present disclosure;
- (24) FIG. 23 is a schematic diagram showing the cat stroking door and the viewing window of the drying machine at still another view angle according to an embodiment of the present disclosure;
- (25) FIG. 24 is an enlarged view of part J circled in FIG. 23;
- (26) FIG. 25 is an enlarged view of part K circled in FIG. 23;
- (27) FIG. 26 is a schematic diagram showing the cat stroking door and a second mounting portion of the drying machine according to an embodiment of the present disclosure;
- (28) FIG. 27 is an enlarged view of part L circled in FIG. 26;
- (29) FIG. 28 is a schematic diagram of a locking member of the drying machine according to an embodiment of the present disclosure;
- (30) FIG. 29 is a schematic diagram of a fragrance box of the drying machine according to an embodiment of the present disclosure; and
- (31) FIG. 30 is an enlarged view of part M circled in FIG. 29.

LIST OF REFERENCE NUMERALS

(32) **100**: Drying machine; **1**: Box body; **11**: Drying cavity; **12**: First accommodation cavity; **13**: Rear wall; **131**: Second air inlet; **14**: Second accommodation cavity; **15**: Extension; **151**: First concave surface; **16**: Dust filter screen; **17**: Opening; **171**: Cat stroking opening; **1711**: Avoidance opening; **1712**: Ventilation hole; **172**: Cat stroking door; **173**: Snap-fitting section; **1731**: First stopping protrusion; **174**: Snap-fitting groove; **175**: Mounting opening; **18**: Limiting groove; **19**: Placement slot; **21**: Fan wheel; **22**: Electric motor; **23**: Fan wheel housing; **231**: Air channel; **2311**: Air channel outlet; **232**: First partition plate; **2321**: First air inlet; **2322**: First fan wheel housing; **233**: Electric motor support; **2331**: Second fan wheel housing; **3**: Second partition plate; **31**: Air outlet; **311**: First air outlet orifice; **312**: Second air outlet orifice; **32**: Groove; **33**: Mounting hole; **34**: Partition plate body; **341**: Boss; **35**: Partition plate section; **4**: Snap-fitting assembly; **41**: Fastener; **411**: Snap portion; **412**: Limiting protrusion; **413**: Penetrating hole; **414**: Second concave surface; **42**: Slot; **5**: Supporting member; **6**: Mounting assembly; **61**: Sliding entrance; **62**: First mounting recess; **621**: First rib; **622**: Second rib; **7**: First limiting block; **8**: Door body; **9**: Pivoting assembly; **91**: First mounting portion; **911**: Mounting lug; **92**: Second mounting portion; **93**: Pivot shaft; **94**: Torsion spring; **10**: Viewing window; **101**: Flange section; **102**: Second stopping protrusion; **103**: Second limiting block; **20**: Locking assembly; **201**: Locking slot; **202**: Locking member; **203**: Locking protrusion; **2031**: Bulge; **30**: Barrier plate; **301**: Second mounting recess; **40**: Fragrance box; **401**: Shell; **4011**: Third mounting recess; **4012**: Protrusion; **4013**: Shell side wall; **4014**: Matching groove; **4015**: Through hole; **4016**: Third rib; **402**: Cover; **4021**: Cover side wall; **4022**: Matching protrusion; **50**: Magnetic attraction assembly; **501**: Screw; **502**: Magnetic member; **60**: Seal; **70**: Upper housing; **701**: First mounting cavity; **702**: Second mounting cavity; **80**: Heating assembly; **90**: Electrical control element.

DETAILED DESCRIPTION OF THE EMBODIMENTS

(33) The embodiments of the present disclosure will be described below in detail, examples of which are illustrated in the accompanying drawings, wherein like or similar reference numerals refer to the same or similar elements or elements having the same or similar functions throughout the description. The embodiments described below with reference to accompanying drawings are exemplary and are intended to be illustrative of the present disclosure, and will not be interpreted as limiting the present disclosure.

(34) In the description, it should be understood that the orientation or positional relationships indicated by the terms “center”, “upper”, “lower”, “front”, “rear”, “left”, “right”, “vertical”, “horizontal”, “top”, “bottom”, “inner”, “outer”, etc. are based on the orientation or positional relationship shown in the accompanying drawings and are only for facilitating the description of the embodiments of the present disclosure and simplifying the description, rather than indicating or implying that a device or an element referred to must have a particular orientation or be constructed and operated in a particular orientation, and therefore will not be interpreted as limiting the embodiments of the present disclosure.

(35) It should be noted that the terms “first” and “second” are used for descriptive purposes only, and cannot be construed as indicating or implying relative importance or implicitly indicating the number of technical features indicated. Thus, the features defined with “first” and “second” can explicitly or implicitly include one or more of the features. Further, in the description of the present disclosure, the term “a plurality of” means two or more, unless otherwise specified.

(36) A drying machine **100** according to an embodiment of the present disclosure will be described below with reference to FIGS. **1-30**. The drying machine **100** may be, but is not limited to, a pet drying machine. In the description of the present application, a pet drying machine is taken as an example to illustrate the drying machine **100**.

(37) As shown in FIGS. **1-30**, the drying machine **100** according to the embodiment of the present disclosure comprises a box body **1**.

(38) Specifically, a first partition plate **232** is provided in the box body **1**, the first partition plate **232** is located in front of a rear wall **13** of the box body **1**, and a drying cavity **11** and a first accommodation cavity **12** are respectively located on two sides of the first partition plate **232**; the first accommodation cavity **12** is defined between the first partition plate **232** and the rear wall **13**, a fan wheel **21** is provided in the first accommodation cavity **12**, and a rotation axis of the fan wheel **21** is perpendicular to the first partition plate **232** and the rear wall **13**; a first air inlet **2321** is formed in the first partition plate **232**, a second air inlet **131** is formed in the rear wall **13**, and the first air inlet **2321** and the second air inlet **131** each are axially opposite to the fan wheel **21**.

(39) When the fan wheel **21** rotates, an airflow inside the drying cavity **11** is suitable for entering the first accommodation cavity **12** through the first air inlet **2321** under the action of the fan wheel **21** and then flowing into the drying cavity **11**, and an airflow outside the drying machine **100** is suitable for entering the first accommodation cavity **12** through the second air inlet **131** under the action of the fan wheel **21** and then flowing into the drying cavity **11**, so as to realize inner circulation and outer circulation of air by means of one fan wheel **21**.

(40) For example, in the examples shown in FIGS. **1, 4, 5** and **7**, the box body **1** may be a square box body, the fan wheel **21** is provided in an upper portion of the first accommodation cavity **12**, and the first air inlet **2321** and the second air inlet **131** each are axially opposite to the fan wheel **21**. When a pet is dried in the drying machine **100**, the fan wheel **21** rotates, the airflow in the drying cavity **11** in this case may axially enter the fan wheel **21** through the first air inlet **2321**, and meanwhile, the airflow outside the drying machine **100** such as the pet drying machine can axially enter the fan wheel **21** through the second air inlet **131**, and the airflow in the fan wheel **21** can be then expelled to the first accommodation cavity **12** in a radial direction of the fan wheel **21** and be compressed in the first accommodation cavity **12**, and finally flows into the drying cavity **11** under the action of the fan wheel **21** to dry the pet. Thus, compared with a conventional pet drying box,

the present application can simultaneously realize the inner circulation and outer circulation of air by means of one fan wheel **21**, replenish fresh air, and make the pet more safe; also, the supply air rate of the drying machine **100** is increased, the loss of wind energy is reduced, and accordingly the drying efficiency of the drying machine **100** can be improved. In addition, with one fan wheel **21**, it is possible to avoid the nonuniformity of hot and cold air in the box body **1** caused by an additional fan wheel **21**, and it is also possible to effectively reduce noise sources, which reduces the noise during operation of the drying machine **100** and reduces the production cost of the drying machine **100**.

(41) Providing the fan wheel **21** in the first accommodation cavity **12** between the rear wall **13** of the box body **1** and the first partition plate **232** allows the fan wheel **21** to smoothly introduce the airflow outside the drying machine **100** when rotating, so as to achieve the exchange of air inside and outside the drying machine **100**, it is possible to prevent liquid leakage at a bottom wall of the box body **1** which causes a potential safety hazard to the fan wheel **21**, the use safety of the drying machine **100** is improved, and accordingly the service life of the drying machine **100** can be ensured.

(42) According to the drying machine **100**, such as the pet drying machine, of the embodiment of the present disclosure, the interior of the box body **1** is divided into the drying cavity **11** and the first accommodation cavity **12** by locating the first partition plate **232** in front of the rear wall **13** of the box body **1**, the first accommodation cavity **12** is defined between the first partition plate **232** and the rear wall **13** of the box body **1**, the fan wheel **21** is provided in the first accommodation cavity **12**, the first air inlet **2321** is formed in the first partition plate **232**, the second air inlet **131** is formed in the rear wall **13**, and the first air inlet **2321** and the second air inlet **131** each are axially opposite to the fan wheel **21**. Thus, the air supply rate of the drying machine **100** is increased, the loss of wind energy is reduced, the drying efficiency of the drying machine **100** is improved, the nonuniformity of the hot and cold air in the box body **1** caused by an additional fan wheel **21** can also be avoided, and the noise during the operation of the drying machine **100** and the production cost of the drying machine **100** can be reduced.

(43) According to some embodiments of the present disclosure, as shown in FIG. **13**, a fan wheel housing **23** is provided between the first partition plate **232** and the rear wall **13**, the fan wheel **21** is arranged in the fan wheel housing **23**, and an air outlet channel **231** is defined between the fan wheel **21** and the fan wheel housing **23**; when the fan wheel **21** rotates, the airflow inside the drying cavity **11** sequentially flows through the first air inlet **2321** and the air channel **231** and then flows into the drying cavity **11**, and the airflow outside the drying machine **100** sequentially flows through the second air inlet **131** and the air channel **231** and then flows into the drying cavity **11**. When the fan wheel **21** rotates, the fan wheel **21** can radially expel the airflow flowing toward the fan wheel **21** into the air outlet channel **231** and compress the airflow in the air outlet channel **231**, and the compressed airflow can flow toward the drying cavity **11** under the guiding effect of the air outlet channel **231**. As a result, it is possible to avoid the loss of air supply rate while allowing the airflow flowing toward the fan wheel **21** to flow into the drying cavity **11** as much as possible, so that the drying efficiency of the drying machine **100** can be improved.

(44) According to some embodiments of the present disclosure, with reference to FIGS. **1**, **3**, **5** and **7**, a second partition plate **3** is provided in the box body **1**, a second accommodation cavity **14** in direct communication with an air channel outlet **2311** of the air channel **231** is defined between the second partition plate **3** and an inner wall of the box body **1** at a position in the fan wheel housing **23** corresponding to the air channel outlet **2311**, an air outlet **31** is formed in the second partition plate **3**, and the drying cavity **11** is located above the second partition plate **3**; and when the fan wheel **21** operates to rotate, the airflow inside the drying cavity **11** is suitable for sequentially flowing through the first air inlet **2321**, the air channel **231** and the second accommodation cavity **14** and flowing into the drying cavity **11** through the air outlet **31**, and the airflow outside the drying machine **100** is suitable for sequentially flowing through the second air inlet **131**, the air

channel **231** and the second accommodation cavity **14** and flowing into the drying cavity **11** through the air outlet **31**. Thus, the airflow flowing from the air channel outlet **2311** can directly flow toward the second accommodation cavity **14**, so that the drying efficiency can be improved. Furthermore, when the pet is lying on the second partition plate **3**, the airflow inside the second accommodation cavity **14** can flow to the pet position through the air outlet **31** so as to dry the pet, and the airflow for drying in this case can be blown to the abdomen of the pet, shortening the action time of the drying process while improving the ventilation effect of the box body **1**, further improving the drying efficiency, and ensuring the drying effect. Also, the second partition plate **3** can allow the water or impurities on the pet to flow onto the bottom wall of the box body **1** through the air outlet **31** under the effect of gravity, ensuring the cleanliness of the drying cavity **11** of the box body **1**.

(45) Further, referring to FIGS. **11-17** in combination with FIG. **1**, the second partition plate **3** is removably arranged in a lower portion of the inside of the box body **1** to facilitate the separation of the second partition plate **3** from the box body **1**, thereby facilitating the cleaning of the bottom wall of the box body **1** and the second partition plate **3**, preventing the accumulation of water and messy environment in the box body **1** to provide clean drying environment, which in turn improves the comfort of the pet during drying. Moreover, after the second partition plate **3** is damaged, the second partition plate **3** can be dismounted from the box body **1** so as to replace the second partition plate **3** with a new one, and the second partition plate is easy to dismount and can prolong the service life of the drying machine **100**.

(46) Still further, the second partition plate **3** is detachably connected to the box body **1** by means of at least one snap-fitting assembly **4**, so as to facilitate the detachable fit between the second partition plate **3** and the box body **1**, meanwhile, the difficulty in mounting and dismounting the second partition plate **3** can be reduced, and the efficiency of dismounting and dismounting can be improved. In such an arrangement, a user can easily dismount the second partition plate **3** so as to clean the bottom wall of the box body **1**, thereby improving the user experience. Of course, in other embodiments, the detachable connection between the second partition plate **3** and the box body **1** can also be implemented by means of hole-post fitting, insertion fitting, etc., which will not be limited herein.

(47) The snap-fitting assembly **4** comprises a snap fastener **41** and a slot **42**, a groove **32** is provided on the second partition plate **3**, the snap fastener **41** is arranged on an inner wall of the groove **32**, and the snap fastener **41** has a snap portion **411**. An extension **15** extending toward the snap fastener **41** is provided on the box body **1**, the slot **42** is defined between the extension **15** and the inner wall of the groove **32**, and the snap portion **411** is fitted in the slot **42**. As shown in FIGS. **11-17**, an opening of the groove **32** may be open into the box body **1** so as to fit with the slot **42** formed in the box body **1**. The slot **42** may be defined between the extension **15** and a bottom wall of the groove **32**. Thus, by means of the fit between the space defined by the extension **15** and the groove **32** and the snap portion **411**, the snap-fit connection between the second partition plate **3** and the box body **1** is implemented, so that the user can easily dismount the second partition plate **3** to clean the bottom of the box body **1**, facilitating cleaning and replacement, further improving the comfort of the pet during drying, and improving the user experience. Optionally, the snap fastener **41** and the inner wall of the groove **32** may be formed as one piece, so that the structural reliability of the snap fastener **41** can be enhanced.

(48) In some optional embodiments, the cross section of the snap fastener **41** is generally U-shaped, and at least a portion of two opposite sides of the U-shaped snap fastener may be spaced apart. With such an arrangement, when being pressed by the user, the two sides of the U-shaped snap fastener can get close to each other to facilitate separation of the snap fastener **41** from the slot **42**, so that the snap fastener **41** can be disengaged from the slot **42**. Of course, the cross section of the snap fastener **41** may also have another shape, which is not limited herein.

(49) One end of the snap fastener **41** is connected to the inner wall of the groove **32**, the other end

of the snap fastener **41** has the snap portion **411**, a limiting protrusion **412** is provided on a side of the snap fastener **41** facing the extension **15**, and the limiting protrusion **412** and the snap portion **411** are respectively located on two sides of the extension **15** in a thickness direction (for example, in a vertical direction as shown in FIG. **17**). Thus, during actual mounting, it is possible to snap-fit the extension **15** between the limiting protrusion **412** and the snap portion **411** to prevent the second partition plate **3** from movement, which in turn enhances the mounting stability of the second partition plate **3** and the box body **1**.

(50) Further, as shown in FIG. **15**, a side surface of the limiting protrusion **412** facing a side wall of the box body **1** obliquely extends toward the side wall of the box body **1** from bottom to top, and the thickness of the limiting protrusion **412** gradually increases from an end close to the bottom of the box body **1** to the end away from the bottom of the box body **1**, so that a surface of the side of the limiting protrusion **412** facing the side wall of the box body **1** is formed as a bevel. Thus, on the one hand, a mutually abutting area between the extension **15** and the limiting protrusion **412** can be increased, and on the other hand, the space occupied by the limiting protrusion **412** can be decreased, so as to facilitate the disengagement of the snap fastener **41** together with the limiting protrusion **412** from the slot **42**. Further, an end surface of the end of the limiting protrusion **412** away from the bottom of the box body **1** is adapted to the bottom of the extension **15**, so as to ensure the mounting reliability of the extension **15** and the limiting protrusion **412**. Optionally, the limiting protrusion **412** and the snap fastener **41** may be formed as one piece to improve the structural reliability.

(51) Further, part of an upper surface of the extension **15** is formed as a first concave surface **151** recessed downwards. When the snap fastener **41** and the slot **42** are mounted or separated, the user can abut against the first concave surface **151** with his/her hand so as to improve the user's comfort during mounting or dismounting. For example, the first concave surface **151** may be an arc-shaped surface, so that the comfort can be further improved.

(52) In some optional embodiments, as shown in FIG. **18**, a through penetrating hole **413** is formed in the snap portion **411** to decrease the weight of the snap portion **411** and to achieve a lightweight design of the snap-fitting assembly **4**, and it is conducive for the snap portion **411** to deform toward the penetrating hole **413** when the user presses the snap portion **411** to separate the snap portion **411** from the slot **42**, so that the force of the user required for pressing the snap portion **411** can be decreased by virtue of the penetrating hole **413**, facilitating quick mounting and dismounting of the second partition plate **3** by the user.

(53) Referring to FIG. **16**, at least part of the side surface of the snap portion **411** facing the side wall of the box body **1** is formed as a second concave surface **414** recessed in a direction away from the side wall of the box body **1**, wherein the second concave surface **414** may be designed as another shape according to the requirement of the user, for example, the second concave surface **414** may be a circular arc surface, so that the hand of the user can fit the second concave surface **414** when the user holds it, increasing the contact area between the two, and thus the user can press the snap portion **411** with a small force, enhancing the practicability of the snap portion **411**.

(54) Optionally, the snap fastener **41** may be an elastic fastener, and the snap fastener **41** arranged in such a way that the snap fastener can elastically deform when the user applies a force to the snap fastener **41**, so as to facilitate the separation of the snap fastener **41** from the slot **42** when the user presses the snap fastener **41**, thereby implementing the dismounting or mounting of the two. In addition, the elastic member has a good recovery performance, and when the user stops applying the force to the snap fastener **41**, the snap fastener **41** can recover to its shape before pressing by virtue of its own recovery ability, so as to facilitate the mounting and fit of the snap fastener **41** and the slot **42** and provide a good mounting reliability.

(55) In some optional embodiments, the snap-fitting assembly **4** is located in the front of the box body **1**. For example, when the user opens the box body **1**, the mounting or dismounting of the second partition plate **3** and the box body **1** can be completed by mounting or dismounting the snap

fastener **41** and the slot **42** such that the effect of improving the dismounting convenience can be provided, and the mounting and dismounting efficiency is improved.

(56) Optionally, a plurality of snap-fitting assemblies **4** may be provided, and the plurality of snap-fitting assemblies **4** are spaced apart from each other. For example, two snap-fitting assemblies **4** may be provided, and the two snap-fitting assemblies **4** may be respectively provided on two sides of the box body **1** in a width direction (for example, the left-right direction as shown in FIG. **1**), and during mounting or dismounting, the user may hold the two snap-fitting assemblies **4** respectively with his/her left and right hands, and accordingly can simultaneously mount or dismount the second partition plate **3** from the two sides of the second partition plate **3**. The difficulty in mounting and dismounting the second partition plate **3** is reduced, and the detachable connection between the second partition plate **3** and the box body **1** can be firmly achieved.

(57) Optionally, bulges for increasing a frictional force may be additionally provided on the second partition plate **3**, and may allow the pet to be more stable on the second partition plate **3** for drying. The bulges may be elongated or square, which is not limited herein.

(58) In some optional embodiments, the spacing between the second partition plate **3** and the bottom wall of the box body **1** is between 1-2 cm (including an endpoint value). It should be noted herein that the "spacing" refers to the distance between the bottom surface of the second partition plate **3** and the top of the bottom wall of the box body **1**. The spacing between the second partition plate **3** and the bottom wall of the box body **1** may be 1 cm, or the spacing between the second partition plate **3** and the bottom wall of the box body **1** may also be 2 cm. With such an arrangement, the drying cavity **11** and the second accommodation cavity **14** can have a more reasonable height, an activity space for the pet can also be ensured, the ventilation in the box body **1** can be facilitated, and the fluidity of the airflow inside the box body **1** can be improved, and accordingly the drying efficiency can be improved.

(59) According to some embodiments of the present disclosure, as shown in FIGS. **15-17**, the bottom wall inside the box body **1** is provided with at least one supporting member **5**, and the supporting member **5** can support the second partition plate **3**. For example, a plurality of supporting members **5** may be provided, and the plurality of supporting members **5** may be respectively provided at an edge and the center of the second partition plate **3**, wherein the supporting members **5** provided at the edge of the second partition plate **3** may be symmetrically arranged so as to ensure the supporting effect on the second partition plate **3** and the mounting stability of the second partition plate **3**. Since a certain distance between the second partition plate **3** and the bottom wall of the box body **1** is provided, that is to say, the second partition plate **3** is considered to be suspended inside the box body **1**, the second partition plate **3** has a certain weight itself and the pet will lie on the second partition plate **3**, the second partition plate **3** and the pet on the second partition plate **3** can be well supported by the supporting members **5** so as to ensure a good drying effect.

(60) For example, four supporting members **5** may be provided, the four supporting members **5** are distributed at a distance from each other; one end of each of the supporting members **5** is connected to the bottom wall of the box body **1**, and the other end of the supporting member **5** can be supported on the second partition plate **3** to facilitate the connection between the second accommodation cavity **14** and the first accommodation cavity **12**, so that the airflow can flow to the second accommodation cavity **14** through the first accommodation cavity **12** and enter the drying cavity **11** through the air outlet **31** for drying the pet; and the supporting members **5** dispersedly arranged do not interfere with the flow of the airflow and do not hinder the flow of the air. When the pet lies on the second partition plate **3**, the air can dry the abdomen and other parts of the pet by means of the air outlet **31** in the second partition plate **3** to improve the drying efficiency.

Optionally, the supporting member **5** may be a supporting frame or a supporting post.

(61) Optionally, the second partition plate **3** may also be detachably connected to the supporting member **5** by means of the snap-fitting assembly **4**. The snap fastener **41** of the snap-fitting

assembly **4** is arranged on the second partition plate **3**, and the slot **42** of the snap-fitting assembly **4** is formed in the supporting member **5**. During mounting, the snap fastener **41** of the second partition plate **3** may be snap-fitted into the slot **42** in the supporting member **5** to implement the detachable connection between the second partition plate **3** and the box body **1**.

(62) Further, referring to FIG. **16**, a mounting hole **33** may be formed in the second partition plate **3**, and an end portion of the supporting member **5** is fitted in the mounting hole **33**. Thus, on one hand, the mounting hole **33** can provide a positioning effect so as to facilitate the mounting of the supporting member **5** and the second partition plate **3**, improving the mounting efficiency; on the other hand, the insertion fitting of the supporting member **5** in the mounting hole **33** may improve the supporting stability of the supporting member **5** to ensure the comfort of the pet during drying.

(63) Optionally, a plurality of mounting holes **33** may be provided, the plurality of mounting holes **33** in one-to-one correspondence with the plurality of supporting members **5**, and the plurality of mounting holes **33** are spaced apart from each other. For example, four mounting holes **33** may be provided, and the four mounting holes **33** in one-to-one correspondence with the four supporting members **5**. Thus, when the second partition plate **3** is mounted, the mounting holes **33** can be directly aligned to upper ends of the supporting members **5** so that the upper ends of the supporting members **5** at least partially extend into the mounting holes **33** to prevent the separation of the second partition plate **3** from the supporting members **5**, and the connecting stability of the second partition plate **3** and the supporting members **5** can be enhanced. Further, the second partition plate **3** is supported by the supporting members **5**, so that the second partition plate **3** will not fall off when the pet is in the drying cavity **11**, thereby enhancing the overall structural strength of the drying machine **100**.

(64) According to some embodiments of the present disclosure, as shown in FIG. **17**, a mounting assembly **6** for mounting the second partition plate **3** is provided inside the box body **1**, so that the second partition plate **3** can be prevented from being separated from the box body **1**, which may in turn enhance the structural stability of the second partition plate **3**. Further, as shown in FIG. **19**, the mounting assembly **6** may comprise a sliding entrance **61** and a first mounting recess **62**, and the second partition plate **3** may slide, in the box body **1**, into the first mounting recess **62** along the sliding entrance **61**. Thus, it is possible to reduce the difficulty in mounting and dismounting the second partition plate **3** and the box body **1**, and to enhance the structural stability of the second partition plate **3**. For example, the sliding entrance **61** may be a stop opening. The first mounting recess **62** may be an elongated groove. Two sliding entrances **61** may be provided, the two sliding entrances **61** may be symmetrically provided in two opposite inner sides of the box body **1**, and the first mounting recess **62** is provided in the bottom of the box body **1** at a position corresponding to the sliding entrances **61**. Where the first mounting recess **62** is provided in one side of the bottom of the box body **1**, the slot **42** in the snap-fitting assembly **4** may be provided in the other side relative to the first mounting recess **62**.

(65) Specifically, when the second partition plate **3** is mounted on the box body **1**, the second partition plate **3** can slide into the box body **1** along the sliding entrance **61** until the second partition plate reaches the first mounting recess **62**, the second partition plate **3** is then snap-fitted to the box body **1** by means of the snap-fitting assembly **4**, and the mounting assembly **6** further facilitates the entry of the second partition plate **3** into the box body **1**, so that the efficiency of assembling the second partition plate **3** and the box body **1** can be improved.

(66) Further, referring to FIG. **19**, a first rib **621** and a second rib **622** are provided on the rear wall **13** in the box body **1**, a rear end of the first rib **621** and a rear end of the second rib **622** are both connected to the rear wall **13** of the box body **1**, and the second rib **622** is located below the first rib **621**, such that the first mounting recess **62** is defined jointly by a lower side face of the first rib **621**, an upper side face of the second rib **622**, and the rear wall **13** of the box body **1**, and the sliding entrance **61** is formed between a front end of the first rib **621** and a front end of the second rib **622**.

(67) Thus, during actual mounting, it is easy for the second partition plate **3** to slide into the box body **1** along the sliding entrance **61** until one end of the second partition plate **3** extends into the first mounting recess **62**, and the second partition plate **3** can be limited by means of the first mounting recess **62**, so that the second partition plate **3** is prevented from wagging or displacing, and the structural stability of the second partition plate **3** is enhanced.

(68) Further, the width of the second rib **622** in a front-rear direction (e.g., the dimension in the front-rear direction as shown in FIG. **17**) is greater than the width of the first rib **621** in the front-rear direction. Thus, when the second partition plate **3** is mounted in the first mounting recess **62**, the contact area between the second rib **622** and the second partition plate **3** is increased to ensure the stability of supporting the second partition plate **3**. Also, the second partition plate can be easy to mount and dismount, and the user can quickly mount and dismount the second partition plate **3**, without influence of the first rib **621** on the mounting and dismounting of the second partition plate **3**, which facilitates reducing difficulty in mounting and dismounting the second partition plate **3**, improving the assembly efficiency of the drying machine **100**.

(69) In some optional embodiments, a limiting member for limiting the movement of the second partition plate **3** is provided in the box body **1**. Thus, providing the limiting member allows to well fix the second partition plate **3** to prevent the second partition plate **3** from shifting, it can be ensured that the pet on the second partition plate **3** is effectively dried, the drying efficiency can be improved, and meanwhile, it can also be ensured that there is no play or movement between the second partition plate **3** and the box body **1** when the pet moves on the second partition plate **3**.

(70) Further, the limiting member may be a limiting hole, a first limiting block **7** which can be snap-fitted in the limiting hole is provided on the second partition plate **3**, and the limiting hole and the first limiting block **7** can be snap-fitted or insertion-fitted to ensure the mounting stability of the second partition plate **3** and the box body **1**. For example, the limiting hole may be a rectangular hole, and the first limiting block **7** may be a fastener that can be snapped into the small rectangular hole. In an embodiment of the present disclosure, four limiting holes may be provided, and the four limiting holes are respectively symmetrically provided in two opposite inner sides of the box body **1**, four first limiting blocks **7** are also provided on the second partition plate **3**, and the four first limiting blocks **7** are respectively symmetrically provided on the two opposite sides of the second partition plate **3**. Thus, the first limiting blocks **7** on the second partition plate **3** are snap-fitted in the limiting holes in the box body **1** to allow to better fix the second partition plate **3** and prevent the second partition plate **3** from shifting, ensuring that the pet on the second partition plate **3** is effectively dried and improving the drying efficiency.

(71) According to some embodiments of the present disclosure, the air outlet **31** comprises a plurality of air outlet orifices. The second partition plate **3** comprises a partition plate body **34** and two partition plate sections **35**, wherein the two partition plate sections **35** are respectively arranged on the left side and the right side of the partition plate body **34**, the two partition plate sections **35** extend obliquely upwards from bottom to top in a direction away from the center of the partition plate body **34**, and a plurality of air outlet orifices are formed on both the two partition plate sections **35** and the partition plate body **34**. Referring to FIGS. **1**, **3** and **11**, the second partition plate **3** may be a U-shaped partition plate, each of the partition plate sections **35** may be an arc-shaped partition plate protruding away from the partition plate body **34**, a plurality of air outlet orifices may be distributed over the U-shaped partition plate, the pet may be placed on the partition plate body **34** of the U-shaped partition plate, and an airflow may be jetted from bottom to top through the plurality of air outlet orifices in the partition plate body **34**, so as to facilitate drying the abdomen of the pet. The plurality of air outlet orifices of each of the partition plate sections **35** are arranged at a distance from each other in the length direction of the partition plate section **35**, and the airflow is jetted obliquely upwards through the plurality of air outlet orifices in the two partition plate sections **35** to facilitate drying the back or other parts of the pet, thereby drying the fur of the pet all around, which may in turn improve the utilization rate of wind energy of the

drying machine **100** and provide a high drying speed and a good drying effect. Providing the two partition plate sections **35** to extend obliquely upwards from the bottom to the top in the direction away from the center of the partition plate body **34** allows to direct the airflow in the second accommodation cavity **14** and to reduce a wind resistance to the airflow in the second accommodation cavity **14** so that the airflow can be rapidly blown into the drying cavity **11**. It should be noted that the second partition plate **3** may be shaped to be as close as possible to the bottom wall of the box body **1** depending on the shape of the bottom wall of the box body **1**, so as to provide better fit between the second partition plate **3** and the box body **1**.

(72) Optionally, the plurality of air outlet orifices in the partition plate body **34** may be square to increase the jetting area for the airflow, and the plurality of air outlet orifices in each partition plate section **35** can be circular to provide an air jetting assistance, and these air outlet orifices can work together to achieve a good drying effect.

(73) In some optional embodiments, a plurality of a plurality of bosses **341** are provided on the partition plate body **34**, the plurality of bosses **341** are arranged to be spaced apart from each other on the upper surface of the partition plate body **34**, and the plurality of air outlet orifices in the partition plate body **34** include a plurality of first air outlet orifices **311** and a plurality of second air outlet orifices **312**, the plurality of first air outlet orifices **311** are formed in the partition plate body **34**, and the plurality of second air outlet orifices **312** are formed in the plurality of bosses **341**.

Thus, the airflow can be blown from the second accommodation cavity **14** to the drying cavity **11** through the plurality of first air outlet orifices **311** and the plurality of second air outlet orifices **312**. In addition, providing the bosses **341** and forming the second air outlet orifices **312** in the bosses **341** enable upper ends of the first air outlet orifices **311** and upper ends of the second air outlet orifices **312** to be located at different levels, so that the first air outlet orifices **311** and the second air outlet orifices **312** can be different in air outflow height to prevent all the air outlet orifices from being completely blocking by the pet placed on the second partition plate **3**, which makes the airflow incapable of sufficiently entering the drying cavity **11**, so that the drying effect on the pet can be effectively ensured.

(74) In some embodiments, a drain hole (not shown) is formed in the bottom wall of the box body **1**. Providing the drain hole makes it possible to conveniently drain the water inside the box body **1** to the outside of the box body **1** so as to be conducive to ensure dry environment inside the box body **1**.

(75) The drying machine **100** may further comprise a stopper (not shown) that is detachably arranged at the above-mentioned drain hole to open or close the drain hole. When the stopper is mounted at the drain hole, the stopper can close the drain hole, and in this case, the water inside the box body **1** cannot be drained through the drain hole. For example, when the pet is placed inside the box body **1** for drying, the stopper may be mounted at the drain hole to close the drain hole so as to prevent the airflow from being dispersed through the drain hole to the outside of the box body **1**, so that the drying efficiency and effect can be ensured.

(76) When the stopper is removed from the drain hole, the drain hole is brought into an opened state, and the water inside the box body **1** can flow from the box body **1** through the drain hole. The stopper is detachably connected to the drain hole, so that the mounting or dismounting of the stopper can be facilitated, the user can select the closed or opened state of the drain hole according to actual requirements, and the use experience of the user can be improved.

(77) According to some embodiments of the present disclosure, a heating assembly **80** is provided at the air channel outlet **2311** of the air channel **231**. With reference to FIG. **4**, the heating assembly **80** may extend in the length direction of the air outlet of the air channel **231**, and the heating assembly **80** may increase the flow rate of the airflow and thus the utilization rate of wind energy, and may also rise the temperature of the airflow inside the second accommodation cavity **14** to achieve heating and drying functions, so that the pet in the drying machine **100** can feel comfortable and will not get sick due to only cold air blown by the drying machine **100**, and the

drying efficiency is also improved. In addition, when the drying machine **100** is used as a cat nest, the drying machine **100** may blow warm air in winter and cool air in summer by cooperatively using the fan wheel **21** and the heating assembly **80**, so that the pet will be more comfortable in the drying machine **100**. Optionally, the heating assembly **80** may be a Positive Temperature Coefficient (PTC) heater.

(78) According to some embodiments of the present disclosure, a negative ion generator is provided on the first partition plate **232**, and an output end of the negative ion generator extends into the first accommodation cavity **12**. Thus, when the negative ion generator operates, negative ions generated by the negative ion generator can flow into the drying cavity **11** through the output end to sterilize and disinfect the airflow in the drying cavity **11**.

(79) An ozone disinfection apparatus is provided on the heating assembly **80**. For example, the ozone disinfection apparatus may be an ozone disinfectant or an ozone generator, and pet clothes or pet toys, etc. may be placed in the box body **1** for ozone disinfection, increasing the functionality of the drying machine **100** and further improving the practicability and utilization rate of the drying machine **100**.

(80) Further, the air channel **231** is a spiral air channel. Referring to FIG. **13**, a projection of the spiral air channel on the first partition plate **232** is in the form of a spiral. Thus, providing the air channel **231** as the spiral air channel allows to increase the air outflow from the air channel outlet **2311**.

(81) Still further, a projection of at least a portion of the fan wheel housing **23** on the first partition plate **232** is in the form of a spiral. For example, in the example of FIG. **13**, the projection of at least the portion of the fan wheel housing **23** on the first partition plate **232** is in the form of a Archimedean spiral. Thus, by virtue of the principle of the Archimedes spiral, the length of the air channel **231** can be shortened, the loss of wind energy is small, it can be ensured that the airflow blown from the air channel **231** can have a certain impact force, so that the drying function of the drying machine **100** can be well realized.

(82) According to some embodiments of the present disclosure, referring to FIG. **13**, the air channel **231** has the air channel outlet **2311**, and the cross-sectional area of at least a portion of the air channel **231** gradually increases toward the air channel outlet **2311** in a circumferential direction of the fan wheel **21**. Thus, the airflow can be forced from a narrow portion to a wide portion in an extension direction of the air channel **231** and finally blown out at the air channel outlet **2311** with a reduced pressure intensity, so that it is possible to ensure the airflow blown out from the air channel outlet **2311** having a certain impact force, and the drying efficiency of the drying machine **100** can thus be improved.

(83) According to some embodiments of the present disclosure, as shown in FIG. **4**, the fan wheel housing **23** is arranged on at least one of the first partition plate **232** and the rear wall **13**. The fan wheel housing **23** may be arranged on one of the first partition plate **232** and the rear wall **13**; and optionally, the fan wheel housing **23** may also be provided on both of the first partition plate **232** and the rear wall **13**. Thus, the drying machine **100** may be made compact in terms of structure.

(84) Further, the drying machine **100** further comprises an electric motor support **233**. The fan wheel housing **23** comprises a first fan wheel housing **2322** and a second fan wheel housing **2331**, wherein the first fan wheel housing **2322** is arranged on the first partition plate **232**, the second fan wheel housing **2331** is arranged on the electric motor support **233**, and the second fan wheel housing **2331** and the first fan wheel housing **2322** jointly define a fan wheel accommodation cavity for accommodating the fan wheel **21**. Referring to FIG. **4**, the electric motor support **233** is arranged between the fan wheel **21** and the rear wall **13**, an electric motor **22** is provided inside the fan wheel accommodation cavity, an output shaft of the electric motor **22** is fixed to the fan wheel **21**, the electric motor **22** is mounted on the second fan wheel housing **2331**, and the second fan wheel housing **2331** can support the electric motor **22**, so that the electric motor **22** is stable and reliable during operation, and the drying effect can thus be improved. The electric motor support

233 may be an annular electric motor support adapted to the first fan wheel housing **2322**, the electric motor **22** may be positioned at an inner ring of the annular electric motor support, and a ventilation hole parallel to the rotation axis of the fan wheel **21** is formed in the annular electric motor support, so that the airflow outside the drying machine **100** may flow toward the fan wheel **21** through the second air inlet **131** and the ventilation hole.

(85) Still further, the electric motor support **233** is formed with an air inlet hole that runs therethrough in an axial direction of the fan wheel **21**, the electric motor support **233** is in sealed connection with the rear wall **13** on a peripheral side of the air inlet hole, and outside air sucked by the fan wheel **21** sequentially passes through the second air inlet **13** and the air inlet holes and enters the fan wheel accommodation cavity. Thus, the outside air can directly enter the fan wheel accommodation cavity through the second air inlet **13** and the air inlet holes of the electric motor support **233**, and the drying efficiency can be improved.

(86) In some optional embodiments, the electric motor support **233** is detachably connected to the first partition plate **232**. Thus, the mounting and dismounting of the electric motor support **233** are facilitated.

(87) In some optional embodiments, referring to FIG. 4, the rear surface of the first partition plate **232** extends backwards from the first fan wheel housing **233**. With such an arrangement, the first fan wheel housing **233** and the first partition plate **232** can be integrally formed, so that the steps of assembling the drying machine **100** can be decreased, and the assembly efficiency of the drying machine **100** can thus be improved.

(88) Optionally, the fan wheel **21** may be a forward-inclined fan wheel that can use a centrifugal force to apply work and that can increase the pressure of the airflow to compress the airflow in the air channel **231**, such that the airflow jetted through the air channel outlet **2311** has a certain impact force so as to better achieve the drying function.

(89) According to some embodiments of the present disclosure, the fan wheel **21** is provided as an axially through fan wheel. For example, a hub of the fan wheel **21** may be formed with a plurality of ventilation holes therethrough, and the plurality of ventilation holes may be arranged at a distance from each other in a circumferential direction of the fan wheel **21**. When the fan wheel **21** operates, the airflows inside and outside the box body **1** can enter the fan wheel **21** in the axial direction of the fan wheel **21**, and are then radially expelled by blades of the fan wheel **21**, such that the airflow accumulates and is compressed in the air channel **231** and flows toward the drying cavity **11** of the box body **1** through the air channel outlet **2311**. The ventilation holes can allow the outside airflow to directly enter the fan wheel **21** and increase the airflow entering the fan wheel **21**, such that the air inside and outside the drying machine **100** can circulate fully, and comfortable drying environment can thus be provided.

(90) In some optional embodiments, a fur filter screen is provided at the first air inlet **2321**, and a dust filter screen **16** is provided at the second air inlet **13**. As shown in FIG. 4, the dust filter screen **16** is arranged on a side of the fan wheel **21** away from the inside of the box body **1**, and the fur filter screen is arranged on a side of the fan wheel **21** close to the inside of the box body **1**. The dust filter screen **16** may be arranged directly at the second air inlet **13** of the rear wall **13** of the box body **1**. When the pet is dried in the drying machine **100**, the fur of the pet tends to shed, and the fur filter screen can filter out the fur in the airflow entering the fan wheel **21** in the box body **1**, ensuring that the fan wheel **21** can operate normally. In addition, since the environment where the drying machine **100** is located may be a crowded place such as a shop and the outdoor, the airflow outside the box body **1** inevitably contains dust, and the dust filter screen **16** can filter out the dust in the airflow outside the box body **1** entering the fan wheel **21** to enable the fan wheel **21** to smoothly achieve the functions of inside and outside air suction and can filter out the impurities in the airflow, such the effective service life of the drying machine **100** can be prolonged, and a drying machine **100** with fresh inside air and comfortable environment can also be provided.

(91) Optionally, in the drying machine **100**, a transition area between the position of the air channel

outlet **2311** of the air channel **231** and the bottom wall of the box body **1** is shaped as a circular arc chamfer and may specifically be shaped as a quadrant. With such an arrangement, transitional air guiding can be effectively smoothed, and noise can be effectively reduced.

(92) According to some embodiments of the present disclosure, at least one cat stroking opening **171** is formed in the box body **1**, and a cat stroking door **172** is provided at the cat stroking opening **171** to open or close the cat stroking opening **171**. With reference to FIGS. **1** and **3**, when the pet is positioned inside the box body **1** for drying, the user can push the cat stroking door **172** toward the inside of the box body **1** or pull open it toward the outside of the box body **1**, so as to open the cat stroking opening **171**, and stretch one hand into the cat stroking opening **171** to stroke the pet inside the box body **1** and calm the pet down, facilitating smooth drying, improving the drying effect, and improving the comfort of the pet during drying. Also, the user can stretch one hand into the cat stroking opening **171** to touch the fur of the pet, so as to determine the degree of drying and to determine whether the drying is completed or additional duration for drying is needed, thereby improving the drying efficiency; and the cat stroking door **172** covers the cat stroking opening **171** to prevent the feet of the pet from stretching out from the cat stroking opening **171**, ensuring that the whole body of the pet is dried. The cat stroking opening **171** has a small open area and may be designed only for a person to stretch or retract his/her hands, so that the pet can be prevented from escaping from the box body **1**.

(93) For example, as shown in FIG. **1**, the user can push the cat stroking door **172** from the outside of the box body **1** to the inside of the box body **1** to open the cat stroking opening **171**. On one hand, with such an arrangement, it is convenient and labor-saving for the user; and on the other hand, the pet can be prevented from opening the cat stroking door **172** in the box body **1** to ensure the drying efficiency.

(94) Further, referring to FIGS. **23** and **24**, the cat stroking door **172** is pivotably connected to the box body **1** by means of a pivoting assembly **9** to facilitate the opening and closing of the cat stroking door **172**. The door is labor-saving and easy to open or close, the connecting stability of the box body **1** and the cat stroking door **172** can also be improved, and the reliability during pivoting of the cat stroking door **172** can be ensured. The pivoting assembly **9** comprises a first mounting portion **91**, a second mounting portion **92** and a pivot shaft **93**, wherein the first mounting portion **91** is arranged on the box body **1**, the second mounting portion **92** is arranged on the cat stroking door **172**, and the pivot shaft **93** penetrates the first mounting portion **91** and the second mounting portion **92** to allow the cat stroking door **172** to be pivotable relative to the box body **1**. Thus, providing the first mounting portion **91** and the second mounting portion **92** allows for ease of the mounting of the pivot shaft **93**, and the stability during pivoting can be improved.

(95) Still further, as shown in FIG. **23**, the drying machine **100** further comprises at least one torsion spring **94**, wherein the torsion spring **94** is sleeved on the pivot shaft **93**, and the torsion spring **94** normally forces the cat stroking door **172** in a closing direction of the cat stroking opening **171**. For example, the torsion spring **94** may have an abutting portion, one end of the abutting portion may abut against the side wall of the box body **1**, and the other end of the torsion spring **94** may abut against the second mounting portion **92**. The pivot shaft **93** may penetrate the torsion spring **94** in the length direction of the torsion spring **94**, and two ends of the pivot shaft are respectively connected to two ends of a mounting seat.

(96) It can be understood that the torsion spring **94** is a mechanical force accumulation structure and has a self-resetting function; the torsion spring **94** has a certain force to fasten the cat stroking door **172** thereon, and a certain force is required for opening the cat stroking door **172**, making it difficult for the pet to open the door by itself, so as to ensure that the whole body of the pet is dried inside the box body **1** and to improve the drying effect. When a person's hands exit the cat stroking opening **171**, the cat stroking door **172** can automatically return by means of the torsion spring **94** to continue to cover the cat stroking opening **171** and to prevent the pet from escaping or other impurities from entering the box body **1**, which may affect the drying effect.

(97) According to some embodiments of the present disclosure, one side of the box body **1** is open, and a door body **8** is arranged on the above-mentioned side of the box body **1** to open or close the above-mentioned side of the box body **1**. For example, a front side of the box body **1** is open, and the door body **8** may be arranged on the front side of the box body **1** to open or close the front side of the box body **1**, so that the user can easily place the pet into the box body **1** or take the pet out of the box body **1**. Alternatively, an upper side of the box body **1** is opened, the door body **8** may be arranged on the upper side of the box body **1** to open or close the upper side of the box body **1** so as to reduce the possibility of escaping of the pet, and a specific position of the door body **8** mounted on the box body **1** can be specifically arranged according to actual requirements, which is not limited herein.

(98) In some optional embodiments, as shown in FIG. **23**, the first mounting portion **91** may comprise two mounting lugs **911** spaced apart from each other, the second mounting portion **92** is located between the two mounting lugs **911**, and the pivot shaft **93** penetrates the two mounting lugs **911** and the second mounting portion **92**, so that the mounting reliability of the pivot shaft **93** and the first mounting portion **91** and of the pivot shaft **93** and the second mounting portion **92** can be ensured, and smooth pivoting of the cat stroking door **172** can be facilitated.

(99) The two mounting portions can limit the position of the torsion spring **94** to prevent the torsion spring **94** from being disengaged from the pivoting assembly **9**, and the pivot shaft **93** can prevent the torsion spring **94** from deformation, such that the connection between the cat stroking door **172** and the box body **1** is more secure and effective for a long time, and the effective service life of the drying machine **100** is ensured.

(100) According to some specific embodiments of the present disclosure, as shown in FIGS. **1** and **3**, an opening **17** is formed in the box body **1**, a removable viewing window **10** is provided at the opening **17**, and the cat stroking opening **171** is formed in the viewing window **10**. Providing the viewing window **10** makes it convenient for the user to observe the internal environment inside the box body **1**, and the user can observe the conditions of the pet without opening the cat stroking door **172** in order to timely watch the mood of the pet, so that a smooth drying process can be provided, the pet can be pleasant, and the practicability of the drying machine **100** can thus be improved. For example, the viewing window **10** may be made of a transparent material to further facilitate viewing of the pet inside the box body **1**. The viewing window **10** may be adapted to the cat stroking door **172** in terms of shape, for example, the viewing window **10** may be ring-shaped.

(101) Further, referring to FIG. **20**, one end (e.g., an upper end as shown in FIG. **1**) of the cat stroking door **172** is pivotably connected to the viewing window **10** by means of the pivoting assembly **9**, and a locking assembly **20** is provided between the other end (e.g., a lower end as shown in FIG. **1**) of the cat stroking door **172** and the viewing window **10**. The cat stroking door **172** is suitable for opening the cat opening **171** when the locking assembly **20** is unlocked, and the cat stroking door **172** is fixed to the viewing window **10** when the other end of the cat stroking door **172** is locked by the locking assembly **20**. Thus, providing the locking assembly **20** allows the cat stroking door **172** to be locked to the viewing window **10**, and the locking assembly **20** needs to be unlocked first when a person wishes to open the cat stroking door **172**, which can prevent the cat stroking door **172** from being opened by accident. The locking assembly **20** may be a locking fastener.

(102) Further, with reference to FIG. **25**, the locking assembly **20** comprises a locking slot **201** and a locking member **202**, wherein the locking slot **201** is formed in the cat stroking door **172**, the locking member **202** is movably arranged on the viewing window **10**, a locking protrusion **203** is provided on the locking member **202**, and the fitting between the locking protrusion **203** and the locking slot **201** can facilitate the adjustment of locked and unlocked states of the locking assembly **20** and can increase the contact area and improve the mounting reliability. When the locking protrusion **203** is fitted in the locking slot **201**, the cat stroking door **172** is fixed to the viewing window **10**, and when the locking protrusion **203** is separated from the locking slot **201**, the cat

stroking door **172** is suitable for opening the cat stroking opening **171** so as to adjust the locked and unlocked states of the locking assembly **20**, and a high stability of fixing of the two in the locked state is provided. For example, at least one of the locking member **202** and the locking slot **201** is made of a deformable material in order that the locking protrusion **203** can move into the locking slot **201** to provide the locked state. As shown in FIG. 25, the locking protrusion **203** is provided with a projection **2031**, and the projection **2031** may protrude from the locking protrusion **203** in the thickness direction of the locking protrusion **203** to facilitate the fitting between the locking protrusion **203** and the locking slot **201**.

(103) In some optional embodiments, as shown in FIG. 27, the cat stroking door **172** is formed with an avoidance opening **1711**, the avoidance opening **1711** is arranged adjacent to the locking slot **201**. When the locking protrusion **203** is separated from the locking slot **201**, the locking protrusion **203** is opposite to the avoidance opening **1711**. Thus, when the locking assembly **20** is in the unlocked state, the locking protrusion **203** can pass through the avoidance opening **1711**, such that the cat stroking door **172** can be pivoted inwards or outwards to open the cat stroking opening **171**, resulting in a structural reliability.

(104) Further, as shown in FIG. 27, the locking slot **201** and the avoidance opening **1711** may be arranged in a circumferential direction of the cat stroking door **172**, and the locking member **202** is movable in the circumferential direction of the cat stroking door **172**. Thus, the locking member **202** may be moved in the circumferential direction of the cat stroking door **172**. For example, the locking member **202** slidably fits with the viewing window **10** to improve the smoothness of movement, improving the use experience of the user.

(105) In some optional embodiments, referring to FIG. 25, a barrier plate **30** extending toward the center of the box body **1** is provided on the viewing window **10**, a second mounting recess **301** is defined between the barrier plate **30** and the viewing window **10**, and the locking member **202** is movably arranged in the second mounting recess **301**. Providing the barrier plate **30** makes it possible to prevent the pet from touching the locking member **202** by accident, which may change the locked or unlocked state of the locking member **202**, so that the use safety of the drying machine **100** can be improved. According to some embodiments of the present disclosure, as shown in FIG. 23, a side at an edge of the opening **17** in a thickness direction is provided with a plurality of snap-fitting sections **173** arranged at a distance from each other in the circumferential direction of the cat stroking door **172**, each snap-fitting section **173** is spaced apart from the edge of the opening **17** to define a snap-fitting groove **174**, and a mounting opening **175** is defined between two adjacent snap-fitting sections **173**. A peripheral edge of the viewing window **10** is provided with a plurality of flange sections **101** arranged at a distance from each other in the circumferential direction of the cat stroking door **172**, and each of the plurality of flange sections **101** passes through a corresponding mounting opening **175** and is rotated relative to the opening **17** to be fitted in a corresponding snap-fitting groove **174**. Thus, it is possible to detachably connect the box body **1** to the viewing window **10**, the design is ingenious, and the efficiency of mounting and dismounting is high, such that the dismounting or mounting of the viewing window **10** can be implemented by rotating the viewing window **10**, and positioning is facilitated. For example, some of the plurality of flange sections **101** may pass through corresponding mounting openings **175** and may be rotated relative to the openings **17** to be fitted in corresponding snap-fitting grooves **174**.

(106) Further, as shown in FIGS. 22 and 23, a first stopping protrusion **1731** is provided at one end of at least one snap-fitting section **173** in the circumferential direction, a second stopping protrusion **102** is provided at one end of at least one flange section **101** in the circumferential direction, the second stopping protrusion **102** cooperates with the first stopping protrusion **1731** to limit continuous rotation of the cat stroking door **172** relative to the opening **17**. During dismounting, when a plurality of flange sections **101** are respectively fitted in a plurality of corresponding snap-fitting grooves **174**, the first stopping protrusion **1731** can abut against the second stopping protrusion **102** to prevent further rotation of the cat stroking door **172** relative to

the box body **1**. Providing the two stopping protrusions allows for an improved accuracy of mounting, and allows to facilitate positioning. For example, one end of the flange section **101** may be fitted in a corresponding snap-fitting groove **174**, and the other end of the flange section **101** may be provided with a second stopping protrusion **102**.

(107) In some examples, referring to FIGS. **2** and **21** in combination with FIGS. **9** and **23**, a limiting groove **18** is formed in the box body **1**, a second limiting block **103** is provided on the viewing window **10**, and when the flange sections **101** are fitted in the snap-fitting grooves **174**, the second limiting block **103** is fitted in the limiting groove **18** to limit the rotation of the cat stroking door **172** relative to the opening **17**. The second limiting block **103** provided on the viewing window **10** can be snap-fitted to the limiting groove **18** and inserted into the limiting groove **18** when rotated together with the viewing window **10**. By means of the fitting between the second limiting block **103** and the limiting groove **18**, the positioning is facilitated, the fixing of the viewing window **10** can be stable on the box body **1**, and the viewing window is not easily damaged by the pet in the box body **1**. For example, the limiting groove **18** and the second limiting block **103** may be located at the upper end of the drying machine **100** to further reduce the possibility of damages to the two caused by the pet. For example, an opening of the limiting groove **18** may face the second limiting block **103**. When the user applies a large acting force, the second limiting block **103** may be forced into the limiting groove **18** in the circumferential direction of the viewing window **10**, achieving the limiting fixation of the second limiting block **103** and the limiting groove **18**.

(108) For example, during mounting, the description is given by taking the direction from the inner side face of the box body **1** to the outer side face of the box body **1** as an example, at the beginning of mounting the viewing window **10**, the plurality of flange sections **101** are located at the corresponding mounting openings **175** in this case, the plurality of flange sections **101** can be fitted in the corresponding snap-fitting grooves **174** by rotating the viewing window **10** counterclockwise, the first limiting block **7** can deform to extend into the limiting groove **18** and then return to its original state, and will not continue moving under the limiting of a side wall of the limiting groove **18**, so as to realize the relative fixing of the two. During dismounting, a clockwise acting force is applied to rotate the viewing window **10** such that the second limiting block **103** exits the limiting groove **18**, and the viewing window **10** can be rotated in the circumferential direction of the opening **17** in this case. When the flange sections **101** completely exit the snap-fitting grooves **174**, the flange sections **101** are located at the corresponding mounting openings **175**, and the viewing window **10** can be dismantled from the box body **1** so as to remove the viewing window **10**. When the first stopping protrusion **1731** abuts against the second stopping protrusion **102**, the viewing window **10** cannot continue rotating.

(109) According to some embodiments of the present disclosure, the wall forming the drying cavity **11** is formed with at least one through ventilation hole **1712**. For example, as shown in FIGS. **4**, **20**, **22** and **23**, a plurality of through ventilation holes **1712** may be formed in the cat stroking door **172**, and the ventilation holes **1712** can facilitate ventilation of the inside of the box body **1** of the drying machine **100**, so that the pet in the box body **1** can breathe fresh air and feel comfortable. Of course, the ventilation holes **1712** may also be formed in the door body **8**. In addition, a nameplate may be provided on the viewing window **10**, texts, patterns, or other contents may be imprinted on the nameplate to facilitate a diversified design of the drying machine **100**.

(110) In some optional embodiments, in combination with FIGS. **1** and **3**, two cat stroking openings **171** are provided, and the two cat stroking openings **171** are respectively formed in a left side wall and a right side wall of the box body **1**. Two cat stroking doors **172** are provided, and the two cat stroking doors **172** are respectively provided at the two cat stroking openings **171** to respectively open or close the corresponding cat stroking openings **171**. Thus, the user can further conveniently observe the pet, and the user can observe or stroke the pet from a plurality of perspectives to improve the experience.

(111) According to some embodiments of the present disclosure, as shown in FIGS. 5-8, a placement slot **19** is formed in the box body **1**, and a fragrance box **40** is removably arranged in the placement slot **19** by means of a magnetic attraction assembly, such that the connection between the fragrance box **40** and the placement slot **19** is relatively simple, facilitating the mounting and dismounting of the fragrance box **40**. The fragrance box **40** has a first configuration and a second configuration in the placement slot **19**, and at least one through hole **4015** is formed in the fragrance box body **19**. When the fragrance box **40** is in the first configuration, the interior of the fragrance box **40** is not in communication with the interior of the box body **1**, and when the fragrance box **40** is in the second configuration, the interior of the fragrance box **40** is in communication with the interior of the box body **1** through the through holes **4015**.

(112) As shown in FIGS. 7 and 8, when it is required to remove the unpleasant odor in the drying machine **100** such as a pet drying machine, the fragrance box **40** may be mounted in the placement slot **19** of the box body **1**, and in this case, the fragrance box **2** is brought into the second configuration, so that the fragrance box **2** can effectively diffuse a fragrance through the through holes **215** to remove the unpleasant odor in the drying machine **100**, and the fragrance box **40** can be firmly fixed to the box body **1**. As shown in FIGS. 5 and 6, when it is not required to remove the unpleasant odor for the drying machine **100**, the fragrance box **40** may be placed in the placement slot **19** in the first configuration, in this case, the fragrance in the fragrance box **40** cannot be diffused into the box body **1**, and the fragrance box **40** is stably mounted in the placement slot **19**, so that the fragrance box **40** can be prevented from being shifted or lost. In addition, when the fragrance in the fragrance box **40** is not strong, the fragrance box **40** can be directly removed from the placement slot **19** and replaced with a new fragrance box **40**.

(113) According to some embodiments of the present disclosure, the through holes **4015** are formed in a side surface of the fragrance box **40**. An inner wall of the placement slot **19** closes the through holes **4015** when the fragrance box **40** is in the first configuration, and the through holes **4015** are located on an open side of the placement slot **19** when the fragrance box **40** is in the second configuration. As shown in FIG. 5, the fragrance box **40** is generally in a rectangular shape, and the through holes **4015** may be formed in the upper or the lower surface of the fragrance box **40**, and may also be formed in four circumferential surfaces of the fragrance box **40**. When it is required to remove the unpleasant odor for the drying machine **100**, a surface of the fragrance box **40** provided with the through holes **4015** may be opposite to the open side of the placement slot **19**, such that the fragrance in the fragrance box **40** can be diffused into the box body **1** through the through holes **4015**; and when it is not required to remove the unpleasant odor for the drying machine **100**, a surface of the fragrance box **40** provided with the through holes **4015** may face away from the open side of the placement slot **19**, and the inner wall of the placement slot **19** opposite to the open side may close the through holes **4015** in this case, so that the fragrance in the fragrance box **40** cannot be diffused.

(114) In some optional embodiments, a seal **60** is provided on the inner wall of the placement slot **19**, and when the fragrance box **40** is in the first configuration, the through holes **4015** are sealed by the seal **60**. Referring to FIGS. 6 and 8, the seal **60** is arranged on the top wall of the placement slot **19**, and the through holes **4015** are sealed by the seal **60** when the fragrance box **40** is in the first configuration. When it is not required to diffuse the fragrance from the fragrance box **40**, the through holes **4015** can be sealed by the seal **60** so as to stop the diffusion of the fragrance from the fragrance box **40**.

(115) Optionally, the seal **60** is a silicone pad. However, it is not limited so. Since the silicone pad is flexible and may deform to an extent, the silicone pad can be attached to the inner wall of the placement slot **19** to seal the through holes **4015**, ensuring the sealability between the seal **60** and the through holes **4015**.

(116) According to some embodiments of the present disclosure, the magnetic attraction assembly **50** comprises screws **501** and magnetic members **502**, wherein the magnetic members **502** are

arranged on the fragrance box **40**, and the magnetic members **502** magnetically fit with the screws **501** to removably arrange the fragrance box **40** in the placement slot **19**. Referring to FIGS. **5-8**, the magnetic members **502** are arranged at positions on the placement slot **19** of the box body **1** corresponding to the screws **501** of the fragrance box **40**. Thus, the fragrance box **40** and the box body **1** are detachably connected by means of the screws **501** and the magnetic members **502**, such that the fragrance box **40** and the box body **1** are directly connected by means of a magnet effect, and no additional tool is required for dismounting, so that the fragrance box **40** is easy and quick to replace. For example, two screws **501** may be provided, the two screws **501** may be spaced apart from each other in a length direction of an upper housing **70**, and the two screws **501** each are in one-to-one correspondence with two magnetic members **502**, and each of the screws **501** may pass through the upper housing **70** to be threadedly connected to the inner wall of the box body **1**, such that the screws **501** may securely fix the upper housing **70** to the inner wall of the box body **1** while the screws **501** magnetically fit with the magnetic members **502**. In addition, when the drying machine **100** is moved, the fragrance box **40** possibly drops from the box body **1** due to the detachable connection between the fragrance box **40** and the box body **1**, and the upper housing **70** provided can protect the fragrance box **40** such that the fragrance box **40** will not easily drop from the box body **1**.

(117) According to some specific embodiments of the present disclosure, the fragrance box **40** comprises a shell **401** and a cover **402**, wherein one side of the shell **401** is open, the cover **402** is arranged on the above-mentioned side of the shell **401**, the cover **402** and the shell **401** jointly define an accommodating space, the magnetic members **502** are arranged in the accommodating space, and the through holes **4015** are formed in the cover **402** and/or the shell **401**. The through holes **4015** may be formed in one of the cover **402** and the shell **401**, or the through holes **4015** may be formed in both the cover **402** and the shell **401**. Referring to FIG. **29**, one side of the shell **401** in the thickness direction is open, the cover **402** is opposite to the open side of the shell **401**, and a perfume may be placed in the accommodating space to diffuse the fragrance inside the drying machine **100**.

(118) Further, as shown in FIG. **29**, a plurality of through holes **4015** are provided. The plurality of through holes **4015** are formed spaced apart from each other in a surface of the bottom wall of the shell **401**, and the surface of the bottom wall of the shell **401** is the lower surface of the fragrance box **40**. When the shell **401** of the fragrance box **40** placed in the placement slot **19** faces the interior of the box body **1** of the drying machine **100**, the fragrance of the perfume in the fragrance box **40** can be diffused into the box body **1** through the plurality of through holes **4015** to make the air inside the box body **1** more pleasant.

(119) Further, the above-mentioned inner wall of the box body **1** and the upper housing **70** jointly define a first mounting cavity **701** and a second mounting cavity **702** that are in communication with each other, the second mounting cavity **702** is located on a side of the first mounting cavity **701** adjacent to a rear wall **13** of the box body **1**, an electrical control element **90** and an electrical wire are provided in the first mounting cavity **701**, one end of the electrical wire is connected to the electrical control element **90**, and the other end of the electrical wire extends into the second mounting cavity **702**. As shown in FIGS. **5** and **7**, the fan wheel **21** of the drying machine **100** is arranged on the rear wall **13** of the box body **1**, the electrical control element **90** may be arranged in an electrical control box, and the electrical control box is located in the first mounting cavity **701**. One end of the electrical wire passes through the electrical control box to be connected to the electrical control element **90**, and the other end of the electrical wire may passes through the second mounting cavity **702** to be connected to the electric motor **22** so as to control ON and OFF of the fan wheel **21**. Thus, the space between the inner wall of the box body **1** and the upper housing **70** is fully used while the wiring of the electrical wire is facilitated, which the drying machine **100** more compact in structure.

(120) Optionally, referring to FIGS. **5** and **7**, the placement slot **19** is formed by a portion of the

upper housing **70** being recessed upwardly. With such an arrangement, the placement slot **19** may be integrally formed with the upper housing **70**, so that the steps of assembling the drying machine **100** can be decreased, and the assembling efficiency of the drying machine **100** can be further improved.

(121) In some optional embodiments, third mounting recesses **4011** are formed in the shell **401**, the magnetic members **502** are arranged in the third mounting recesses **4011**, and the magnetic members **502** may be magnets. For example, in the example of FIG. **29**, two third mounting recesses **4011** may be formed in the shell **401**, and the two third mounting recesses **4011** may be respectively located at two ends in the fragrance box **40**, and two magnets are provided and may be respectively mounted in the two third mounting recesses **4011**. The third mounting recesses **4011** may be shaped to be adapted to the shapes of the magnets to facilitate the mounting of the magnets. Thus, the magnets can be firmly fixed to the shell **401**.

(122) Further, two protrusions **4012** spaced apart from each other are provided on the side wall of the shell **401**, and the third mounting recess **4011** is defined between the two protrusions **4012**. With reference to FIG. **29**, the two protrusions **4012** are respectively arranged on two side walls of the shell **401** in the length direction, the two protrusions **4012** located on the same side wall are spaced apart from each other in the width direction of the shell **401**, and each protrusion **4012** may extend in a direction of the cover **402**, so that the two protrusions **4012** on the same side wall can limit the movement of the magnetic member **502** in the width direction of the shell **401**, further ensuring the stability of the magnetic member **502** in position.

(123) Optionally, referring to FIGS. **29** and **30**, two third ribs **4016** spaced apart are provided on each of mutually opposite side surfaces of two side walls of the shell **401** in the length direction, and each third rib **4016** extends in a direction of the center of the third mounting recess **4011**. Thus, the third ribs **4016** can limit the movement of the magnetic member **502** in the length direction of the shell **401**, such that the magnetic member **502** is stably fixed in the third mounting recess **4011**.

(124) The magnetic members **502** are arranged inside the fragrance box **40**, which is better than the case of arrangement of the magnetic members outside the fragrance box **40**, so that the loss of the magnetic members **502** can be avoided. In addition, the third mounting recesses **4011** are provided in the fragrance box **40**, and the magnetic members **502** can be further fixed inside the fragrance box **40**, such that the fragrance box **40** can be used in combination with the box body **1** for a longer time.

(125) According to some specific embodiments of the present disclosure, as shown in FIG. **29**, the shell **401** has a shell side wall **4013**, an edge of the cover **402** is provided with a cover side wall **4021** extending towards the shell **401**, and the cover side wall **4021** is opposite to the shell side wall **4013**. A plurality of matching protrusions **4022** are provided on one of the cover side wall **4021** and the shell side wall **4013**. A plurality of matching grooves **4014** are formed in the other of the cover side wall **4021** and the shell side wall **4013**, and each matching protrusion **4022** is fitted in a corresponding matching groove **4014**. A plurality of matching protrusions **4022** may be provided on the cover side wall **4021**, and a plurality of matching grooves **4014** (as shown in FIG. **29**) may be formed in the shell side wall **4013**. Alternatively, a plurality of matching grooves **4014** may be provided in the cover side wall **4021**, and a plurality of matching protrusions **4022** may be provided on the shell side wall **4013** (not shown). Thus, the detachable connection of the cover **402** and the shell **401** of the fragrance box **40** can be realized, so that the user can change the perfume in the fragrance box **40** in time, improving the utilization rate of the fragrance box **40**, and meanwhile, the user can also place perfume having his/her favorite fragrance between the shell **401** and the cover **402** according to his/her own preference.

(126) Further, two cover side walls **4021** are provided, the two cover side walls **4021** are respectively arranged on two sides of the cover **402** opposite to each other, and the two cover side walls **4021** each are located in the accommodating space. Two matching protrusions **4022** spaced apart in the length direction of the cover side wall **4021** are provided on each cover side wall **4021**,

and the plurality of matching grooves **4014** are in one-to-one correspondence with the plurality of matching protrusions **4022**. Referring to FIG. 29, the two cover side walls **4021** are respectively two side walls of the cover **402** in the width direction, and the two side walls of the shell **401** in the width direction are shell side walls **4013**. Two matching grooves **4014** are provided in each shell side wall **4013**, and each matching protrusion **4022** is fitted in a corresponding matching groove **4014** so as to detachably connect the cover **402** to the shell **401**. By means of positioning the two cover side walls **4021** in the accommodating space, the internal space of the fragrance box **40** can be effectively utilized, and the width of the fragrance box **40** can be decreased, thereby reducing the mounting space for the fragrance box **40**.

(127) In some optional embodiments, the placement slot **19** is formed at the top of the box body **1**. As shown in FIGS. 5 and 7, the fragrance box **40** is arranged in the placement slot **19** at the top of the box body **1**, and the fragrance box **40** diffuses the fragrance from top to bottom at the top of the box body **1**, so that the interior of the whole box body **1** can be full of the fragrance, making the air inside the box body **1** more pleasant.

(128) In some optional embodiments, arc-shaped grooves are formed on two side walls of the placement slot **11** opposite to each other. With such an arrangement, when the fragrance box **2** is dismounted or the position of the fragrance box **2** is changed, the user can hold the fragrance box **2** through the two arc-shaped grooves and remove the fragrance box **2** from the placement slot **11** to complete the dismounting or position change of the fragrance box **2**.

(129) According to some embodiments of the present disclosure, the placement slot **19** is formed in the rear of the box body **1**. With reference to FIGS. 1, 5 and 7, since the fan wheel **21** of the drying machine **100** is arranged in the rear of the box body **1**, when the drying machine **100** is used for drying the pet, the pet easily moves in the front of the box body **1** by means of positioning the placement slot **19** closer to the rear wall **13** of the box body **1**, so that the pet can be prevented from damaging the fragrance box **40**, thereby prolonging the service life of the fragrance box **40**.

(130) Optionally, an perfume may be provided in the fragrance box **40** in advance, so that the user can directly use the fragrance box **40** without manually filling an perfume during use.

(131) According to some embodiments of the present disclosure, a front side of the box body **1** is open, and the front side of the box body **1** is provided with the door body **8** for opening or closing the above-mentioned front side of the box body **1**. For example, in the examples of FIGS. 1 and 4, when the door body **8** is opened, the pet can be placed into the box body **1** or removed from the box body **1** through the door body **8**. When the pet is to be positioned inside the box body **1** for drying, the door body **8** is opened, the pet can enter the box body **1** through the door body **8**, the door body **8** is then closed, and the fur of the pet is dried in the box body **1** to realize the drying function of the drying machine **100**. After the cat stroking door **172** on the viewing window **10** is removed, the pet such as a cat can freely enter or exit the box body **1** through the cat stroking opening **171**, and the box body **1** has good ventilation and air permeability, so the pet such as the cat can sleep and play in the box body **1** to realize the function of a cat nest. Normally, one pet is dried by the drying machine **100** about once every half a month, and the function of the cat nest is added for the drying machine **100** while the drying function is realized, thereby greatly improving the practicability and utilization rate of the drying machine **100**.

(132) Optionally, multi-stage temperature control and measurement may be provided inside the box body **1** to prevent the discomfort of the pet in the box body **1** due to excess temperature in the box body **1**.

(133) Optionally, an illuminator may be provided inside the box body **1**, such as a LED intelligent lamp, to facilitate mounting or dismounting of the second partition plate **3** in the box body **1** and to facilitate cleaning of the box body **1**.

(134) Other configurations, etc. and operations of the drying machine **100** according to embodiments of the present disclosure are known to those of ordinary skill in the art and will not be described in detail herein.

(135) In the description, the explanation with reference to the terms such as “an embodiment”, “some embodiments”, “exemplary embodiments”, “an example”, “specific examples”, or “some examples” means that specific features, structures, materials, or characteristics described in combination with the embodiment(s) or example(s) are included in at least one embodiment or example of the disclosure. In the description, the illustrative expressions of the above-mentioned terms are not necessarily referring to the same embodiment or example. Moreover, the specific features, structures, materials, or characteristics described herein may be combined in any one or more embodiments or examples in a suitable manner.

(136) Although the embodiments of the present disclosure have been shown and described, those of ordinary skill in the art can understand that various changes, modifications, substitutions, and variations can be made to these embodiments without departing from the principles and spirit of the present disclosure. The scope of the present disclosure is defined by the claims and equivalents thereof.

Claims

1. A drying machine comprising: a box body, wherein a first partition plate is provided in the box body, the first partition plate is located in front of a rear wall of the box body, and a drying cavity and a first accommodation cavity are respectively located on two sides of the first partition plate, the first accommodation cavity is defined between the first partition plate and the rear wall, a fan wheel is provided in the first accommodation cavity, and a rotation axis of the fan wheel is perpendicular to the first partition plate and the rear wall, a first air inlet is formed in the first partition plate, and a second air inlet is formed in the rear wall; when the fan wheel rotates, an airflow inside the drying cavity is suitable for entering the first accommodation cavity through the first air inlet under the action of the fan wheel and then flowing into the drying cavity, and an airflow outside the drying machine is suitable for entering the first accommodation cavity through the second air inlet under the action of the fan wheel and then flowing into the drying cavity to realize the inner circulation and the outer circulation of air by means of one fan wheel, wherein each of the first air inlet and the second air inlet is axially opposite to the fan wheel, a virtual straight line where the rotation axis of the fan wheel is located passes through the first air inlet and the second air inlet, the fan wheel is located between the first air inlet and the second air inlet, an orthographic projection of the fan wheel on the first partition plate overlaps the first air inlet, and an orthographic projection of the fan wheel on the rear wall overlaps the second air inlet.
2. The drying machine according to claim 1, further comprising a fan wheel housing provided between the first partition plate and the rear wall, wherein the fan wheel is arranged in the fan wheel housing, and an air channel is defined between the fan wheel and the fan wheel housing, the fan wheel comprises an air inlet channel located at an inner side of the air channel in a radial direction of the fan wheel, the first air inlet and the second air inlet are directly communicated with the air inlet channel, and when the fan wheel rotates, the airflow in the drying cavity sequentially flows through the first air inlet and the air channel and then flows into the drying cavity, and the airflow outside the drying machine sequentially flows through the second air inlet and the air channel and then flows into the drying cavity.
3. The drying machine according to claim 2, further comprising a second partition plate provided in the box body, wherein a second accommodation cavity in direct communication with an air channel outlet of the air channel is defined between the second partition plate and an inner wall of the box body at a position in the fan wheel housing corresponding to the air channel outlet, an air outlet is formed in the second partition plate, and the drying cavity is located above the second partition plate; and when the fan wheel operates to rotate, the airflow inside the drying cavity is suitable for sequentially flowing through the first air inlet, the air channel, the second accommodation cavity and flowing into the drying cavity through the air outlet, and the airflow outside the drying

machine is suitable for sequentially flowing through the second air inlet, the air channel, the second accommodation cavity and flowing into the drying cavity through the air outlet.

4. The drying machine according to claim 3, wherein the second partition plate is removably arranged in a lower portion of the box body.

5. The drying machine according to claim 3, wherein the air outlet comprises a plurality of air outlet orifices; and the second partition plate comprises: a partition plate body; and two partition plate sections, wherein the two partition plate sections are respectively arranged on the left side and the right side of the partition plate body, the two partition plate sections extend obliquely upwards from bottom to top in a direction away from the center of the partition plate body, and a plurality of air outlet orifices are formed in both the two partition plate sections and the partition plate body.

6. The drying machine according to claim 5, wherein each partition plate section is an arc-shaped partition plate projecting away from the partition plate body.

7. The drying machine according to claim 2, further comprising a heating assembly provided at an air outlet of the air channel.

8. The drying machine according to claim 7, further comprising a negative ion generator provided on the first partition plate, wherein an output end of the negative ion generator extends into the first accommodation cavity; and an ozone disinfection apparatus is provided on the heating assembly.

9. The drying machine according to claim 2, wherein the air channel is a spiral air channel.

10. The drying machine according to claim 2, wherein the air channel has an air channel outlet, and the cross-sectional area of at least a portion of the air channel gradually increases toward the air channel outlet in a circumferential direction of the fan wheel.

11. The drying machine according to claim 2, wherein the fan wheel housing is arranged on at least one of the first partition plate and the rear wall.

12. The drying machine according to claim 11, wherein the drying machine further comprises an electric motor support; and the fan wheel housing comprises: a first fan wheel housing arranged on the first partition plate; and a second fan wheel housing arranged on the electric motor support, the second fan wheel housing and the first fan wheel housing jointly defining a fan wheel accommodation cavity for accommodating the fan wheel.

13. The drying machine according to claim 12, wherein the electric motor support is formed with an air inlet hole that runs therethrough in an axial direction of the fan wheel, the electric motor support is in sealed connection with the rear wall on a peripheral side of the air inlet hole, and outside air sucked by the fan wheel sequentially passes through the second air inlet and the air inlet hole and enters the fan wheel accommodation cavity.

14. The drying machine according to claim 12, wherein the electric motor support is detachably connected to the first partition plate.

15. The drying machine according to claim 12, further comprising an electric motor provided in the fan wheel accommodation cavity and an output shaft of the electric motor fixed to the fan wheel, wherein the electric motor is mounted on the second fan wheel housing.

16. The drying machine according to claim 1, wherein the fan wheel is a forward-inclined fan wheel.

17. The drying machine according to claim 1, further comprising at least one cat stroking opening formed in the box body, and a cat stroking door provided at the cat stroking opening for opening or closing the cat stroking opening.

18. The drying machine according to claim 17, further comprising an opening formed in the box body and a removable viewing window provided at the opening, wherein the cat stroking opening is formed in the viewing window.

19. The drying machine according to claim 18, wherein one end of the cat stroking door is pivotably connected to the viewing window by means of a pivoting assembly; and a locking assembly is provided between the other end of the cat stroking door and the viewing window, the cat stroking door is adapted for opening the cat stroking opening when the locking assembly is

unlocked, and the cat stroking door is fixed to the viewing window when the other end of the cat stroking door is locked by the locking assembly.

20. The drying machine according to claim 1, further comprising a placement slot formed in the box body and a fragrance box removably arranged in the placement slot by means of a magnetic attraction assembly, wherein the fragrance box has a first configuration and a second configuration in the placement slot, at least one through hole is formed in the fragrance box, the interior of the fragrance box is not in communication with the interior of the box body when the fragrance box is in the first configuration, and the interior of the fragrance box is in communication with the interior of the box body through the through hole when the fragrance box is in the second configuration.
